Sanjay Mehta

Computational Biologist & AI Drug Discovery Scientist

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EXECUTIVE SUMMARY

Computational Biologist and Biophysicist bridging Molecular Dynamics (MD) simulations, alchemical free energy perturbation (FEP), and Deep Learning for Structure-Based Drug Design. Specialized in protein-ligand binding kinetics, cryptic allosteric pocket discovery, and generative chemistry for therapeutics.

EDUCATION

Ph.D. in Computational Biology & Biophysics

Massachusetts Institute of Technology (MIT) | 2019 – 2022

Dissertation: Enhanced Sampling Molecular Dynamics for Drug Resistance & Allosteric Pocket Discovery

B.Tech in Biotechnology

Department of Biotechnology & Bioengineering | 2015 – 2019

First Class Honors with Distinction in Bioinformatics & Molecular Genetics

PROFESSIONAL EXPERIENCE

Senior Computational Drug Discovery Researcher — Helix BioTherapeutics Labs

2024 – Present | San Francisco, CA

Leading structure-based drug design and AI-driven lead optimization for novel oncology kinase targets.

- Identified two nanomolar lead candidates confirmed by in vitro biophysical binding assays.
- Scaled automated FEP workflows across 100+ GPU nodes, reducing computation time by 60%.
- Authored 2 peer-reviewed publications on cryptic allosteric pocket discovery.

Molecular Dynamics & Cheminformatics Fellow — BioTech Innovations Group

2022 – 2024 | Boston, MA

Architected high-throughput virtual screening platforms evaluating 2.5M+ compounds using OpenMM and GNN scoring.

- Discovered novel non-covalent inhibitors validated by biophysical surface plasmon resonance (SPR).
- Engineered automated Markov State Model trajectory clustering tool adopted lab-wide.

TECHNICAL SKILLS & EXPERTISE

Molecular Dynamics & Structural Biology: GROMACS, OpenMM, PyMOL, AMBER/CHARMM, AutoDock Vina, Markov State Models
AI & Cheminformatics: RDKit, TorchDrug, DiffDock, Graph Neural Networks (GNNs), AlphaFold / ESMFold, Binding Affinity (DTA)
Genomics & Pipelines: Biopython, CRISPR Off-Target Modeling, RNA-Seq Analysis, MMseqs2 / BLAST, Nextflow / Snakemake
HPC & Scientific Computing: Python, PyTorch, C++ / CUDA, Docker, SLURM HPC, Git & CI/CD